

DeEntropy: An AI Agent for Automated Dimension Reduction and Exploratory Data Analysis

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1 Introduction and System Architecture

Dimension reduction is central to exploratory analysis of high-dimensional biomedical data, but an appropriate workflow depends strongly on the dataset. Decisions such as whether to standardize features, preserve sparse representations, introduce an intermediate linear reduction, or prioritize local versus broader geometric structure are typically made manually. This project develops **DeEntropy**, an automated agent that constructs, executes, evaluates, and refines dimension-reduction workflows for previously unseen datasets with minimal human intervention.

DeEntropy follows an iterative *Inspect* $\rightarrow$ *Plan* $\rightarrow$ *Preprocess* $\rightarrow$ *Execute* $\rightarrow$ *Evaluate* $\rightarrow$ *Reflect* $\rightarrow$ *Replan/Stop* $\rightarrow$ *Report* workflow. An **AnalysisDataset** preserves the matrix, sample and feature identifiers, metadata, and optional labels. The inspector produces a **DatasetProfile** containing dimensions, feature types, missingness, sparsity, constant/near-constant features, scaling characteristics, and count-like structure. The planner combines this profile with a capability registry, feasibility information, computational budgets, previous experiments, available artifacts, and reflection feedback to propose a structured **AnalysisPlan**.

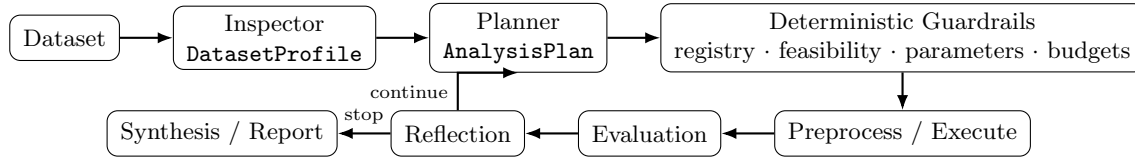


Figure 1: DeEntropy architecture. The decision layer proposes analyses, while deterministic code validates and executes them.

Planning and numerical execution are deliberately separated. Deterministic guardrails validate preprocessing compatibility and ordering, method executability, hyperparameters, graph references and acyclicity, representation compatibility, terminal-experiment budgets, and total analysis-step limits. Only the executor performs numerical analysis.

DeEntropy supports two interchangeable decision providers: a deterministic mock provider for reproducible, network-free execution and a structured OpenAI provider for LLM-based planning and reflection. The finalized experiments reported here use the deterministic provider so that analysis decisions are reproducible. Regardless of provider, proposed plans conform to the same Pydantic schemas and must pass the same deterministic validation before execution.

2 Agent Decision Making

DeEntropy represents an analysis as a directed artifact graph rather than a fixed list of algorithms. A workflow can therefore contain preprocessing, reusable intermediate representations, and multiple terminal embeddings. Intermediate steps do not count as independent terminal experiments and can be reused across branches. This is particularly important for high-dimensional or sparse data, where an expensive representation should not be recomputed for every candidate embedding.

Method selection is profile-aware. The planner considers dimensionality, storage representation, sparsity, scaling, count-like structure, computational feasibility, and registered method strengths and limitations. Hyperparameters are constrained by method-specific rules and dataset dimensions. The system therefore chooses among suitable methods instead of applying every available algorithm indiscriminately.

Terminal embeddings are evaluated using trustworthiness, k -nearest-neighbor (kNN) preservation, and Pearson and Spearman correlations of pairwise distances. The implementation uses scikit-learn for the core PCA/manifold-learning methods and trustworthiness calculation [9, 10], with UMAP provided by the implementation based on McInnes et al. [8].

When labels are available, silhouette score and neighborhood label agreement are computed only as post-hoc external validation; labels are excluded from preprocessing, method selection, hyperparameter selection, and embedding fitting. The reflector uses completed experiments and their metrics to decide whether another strategy is warranted. Final synthesis does not average heterogeneous metrics into a single score. Instead, it separately recommends embeddings for local-neighborhood exploration, broader pairwise structure, and, when available, post-hoc label alignment.

3 Experimental Results

The same DeEntropy architecture was evaluated on two real gene-expression datasets with substantially different representations: the UCI Gene Expression Cancer RNA-Seq dataset, derived from the TCGA PANCAN collection [1], and the PBMC3K count matrix distributed for the Scanpy PBMC3K tutorial [2, 3]. Table 1 summarizes the autonomous workflows.

Table 1: Final workflows selected by DeEntropy.

	Dataset 1: Cancer RNA-seq	Dataset 2: PBMC3K
Samples $\times$ features	801 $\times$ 20,531	2,700 $\times$ 32,738
Storage / labels	Dense / 5 classes	Sparse CSR / none
Key characteristic	$p/n = 25.63$	97.4% sparse, count-like
Preprocessing	Constants $\rightarrow$ variance selection $\rightarrow$ standardization	Constants $\rightarrow$ total normalization $\rightarrow$ log1p $\rightarrow$ variance selection
Reusable intermediate	PCA, 29 components	Truncated SVD, 52 components
Terminal methods	Isomap; Spectral Embedding	UMAP; PCA
Local recommendation	Spectral Embedding	UMAP
Broader structure	Isomap	PCA

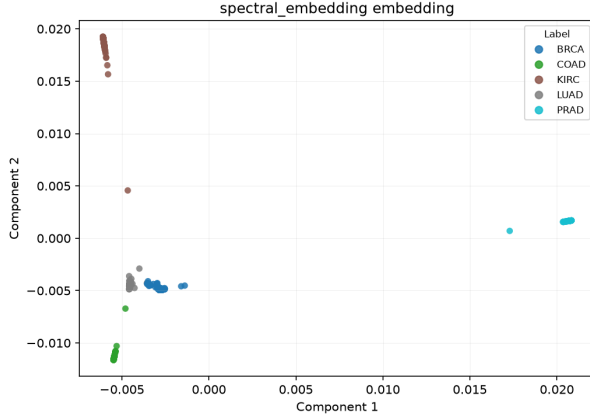
3.1 Dataset 1: Gene Expression Cancer RNA-seq

Dataset 1 contained 801 samples and 20,531 numeric features, including 267 constant and 210 near-constant features. DeEntropy removed the constant features, retained the top 4,056 variance-ranked features, and standardized them. It then constructed a reusable PCA representation with 29 components [4] and evaluated Isomap [5] and Spectral Embedding (Laplacian Eigenmaps) [6] as terminal methods.

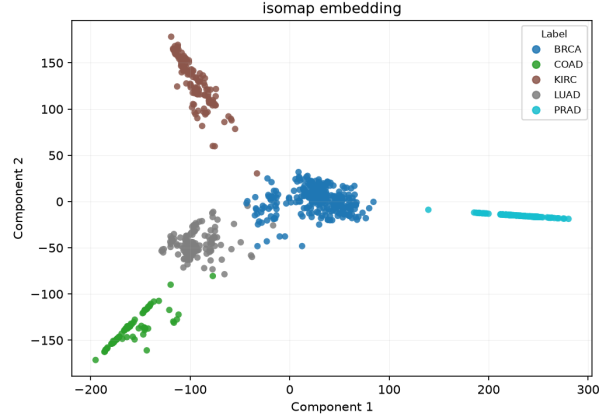
Relative to the PCA-29 parent representation, Spectral Embedding better preserved local structure: trustworthiness was **0.957** and kNN preservation was **0.299**, compared with 0.942 and 0.242 for Isomap. In the separate post-hoc label-based evaluation, Spectral Embedding also showed stronger class alignment (silhouette **0.883**; neighborhood label agreement **0.995**) than Isomap (0.758; 0.989). Isomap better preserved broader pairwise geometry relative to the same parent representation, with Pearson/Spearman distance correlations of **0.753/0.754**, versus 0.680/0.724 for Spectral Embedding. DeEntropy therefore recommended Spectral Embedding for local exploration and post-hoc label alignment, and Isomap as the broader pairwise-structure reference.

3.2 Dataset 2: PBMC3K

Dataset 2 contained 2,700 samples and 32,738 features in a 97.4% sparse, nonnegative count matrix. DeEntropy preserved sparse storage, removed 16,104 constant features, applied generic sample-total normalization,



(a) Spectral Embedding: preferred for local structure and post-hoc class alignment.

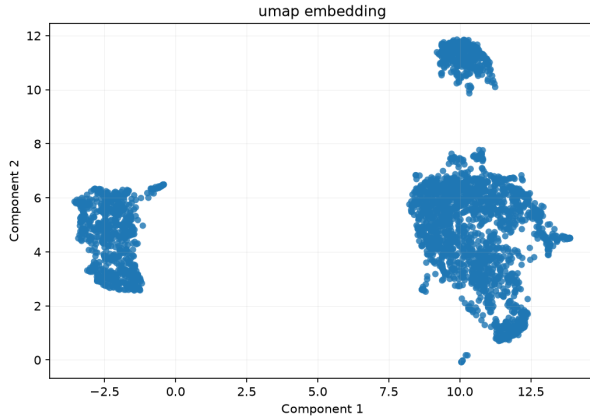


(b) Isomap: preferred as the broader pairwise-structure reference.

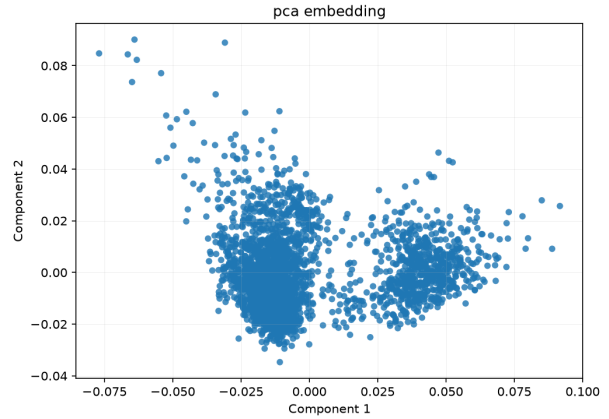
Figure 2: Dataset 1 terminal embeddings. Class labels are used for visualization and post-hoc evaluation only.

followed by sparse-safe `log1p`, and retained 9,402 variance-ranked features. The sparse root was never fully densified. The agent selected noncentered Truncated SVD with 52 components as a reusable intermediate, using a randomized low-rank decomposition [7], followed by UMAP [8] and PCA [4] terminal experiments.

Relative to the shared SVD-52 parent representation, UMAP provided better local preservation, with trustworthiness **0.922** and kNN preservation **0.213**, versus 0.865 and 0.088 for PCA. PCA better preserved broader pairwise relationships, with Pearson/Spearman distance correlations of **0.908/0.890**, versus 0.735/0.786 for UMAP. DeEntropy therefore recommended UMAP for local-neighborhood exploration and PCA as the broader pairwise-structure reference. External validation was unavailable because no class or cell-type labels were provided. Root-fidelity metrics were intentionally omitted because the current evaluator would require densifying the sparse root.



(a) UMAP: preferred for local-neighborhood exploration.



(b) PCA: preferred as the broader pairwise-structure reference.

Figure 3: Dataset 2 terminal embeddings after the shared Truncated SVD representation.

Across both datasets, the results illustrate a central design choice: DeEntropy does not force a single “best” embedding. Local-neighborhood preservation and broader pairwise-structure preservation can favor different methods, so recommendations are tied to the intended exploratory purpose.

4 Strengths and Limitations

A principal strength of DeEntropy is the separation of adaptive decision making from deterministic statistical execution. Dataset characteristics influence preprocessing and method selection, while explicit capability definitions and validation prevent unsupported methods, incompatible representations, invalid hyperparameters, and excessive computational plans from being executed. The artifact graph enables exact intermediate reuse with provenance, and the same architecture successfully handled both a dense labeled dataset and a sparse unlabeled count matrix. Sample identifiers, configurations, analysis histories, metrics, and random seed 42 are retained to support reproducibility.

Important limitations remain. Deterministic mock planning relies on heuristic rules, while real LLM proposals may be uncertain or invalid and therefore require deterministic rejection. The two-terminal-experiment budget allows adaptive comparison but not exhaustive method or hyperparameter search. Diffusion Maps and GPLVM are registered but not executable. Preprocessing is intentionally general rather than domain-specialized: DeEntropy does not perform specialized highly variable gene modeling, clustering, or biological annotation. Sparse-root fidelity is unavailable when evaluation would require full densification, and Dataset 2 lacks labels for external validation. Finally, the reported metrics measure structural preservation and should not be interpreted as biological truth or causality.

5 Toward a General Biomedical Data Analysis Agent

Project 1 is the first stage of a four-project sequence covering (1) dimension reduction and exploratory analysis, (2) generative modeling, (3) multimodality learning, and (4) general AI-agent-based analysis of complex biomedical data. DeEntropy was therefore designed around reusable components—dataset representation, inspection, planning, deterministic validation, execution, evaluation, reflection, and artifact provenance—rather than as an isolated dimension-reduction pipeline.

Future projects will extend the registered analytical capabilities while preserving this control architecture. The next stage will introduce generative modeling tools, followed by multimodal analysis and ultimately coordination of multiple analytical modules within a general biomedical data-analysis agent. A further methodological direction is to incorporate Bayesian reasoning into the decision layer: uncertainty about competing analysis strategies could be represented explicitly and updated using results from completed experiments. Bayesian or LLM-based reasoning would propose and prioritize actions, while deterministic guardrails would remain responsible for validating and executing them.

6 Conclusion

DeEntropy demonstrates that dimension-reduction analysis can be formulated as an iterative agentic decision process rather than a fixed benchmarking pipeline. Across two substantially different biomedical datasets, it autonomously constructed feasible workflows, adapted preprocessing and method selection to the data representation, evaluated competing embeddings, reused intermediate computations, and produced purpose-specific recommendations. The resulting architecture provides a reusable foundation for extending the agent beyond dimension reduction in subsequent projects.

References

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